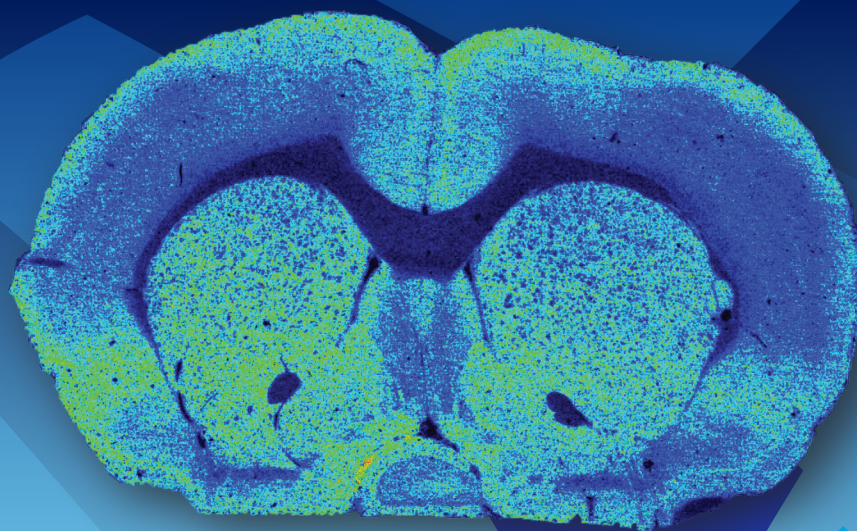
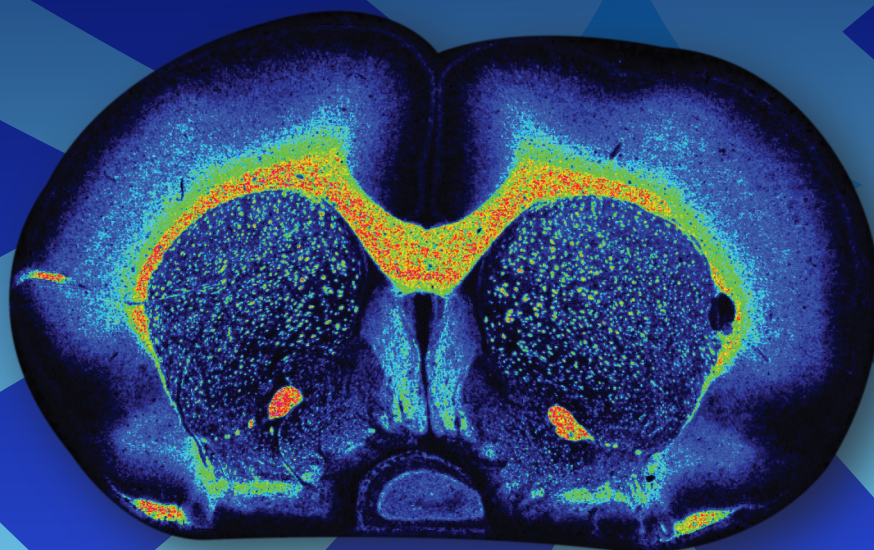




HTX M5 Sprayer™

MALDI Imaging Sample Preparation System

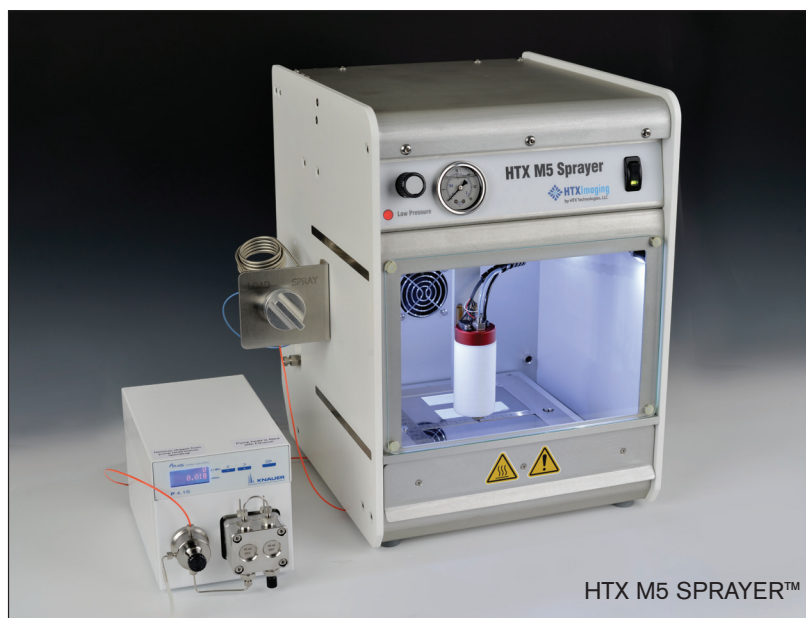


HTXImaging

by HTX Technologies, LLC

HTX M5 Sprayer™ Tissue MALDI Sample Preparation System

HTX M5 Sprayer™
System is an automated
MALDI matrix deposition
system offering
high reproducibility and
superior data quality for
Mass Spectrometry Imaging



The HTX M5 Sprayer™ is an easy-to-use, versatile spraying system that provides automated process for Sample Preparation in Mass Spectrometry Imaging.

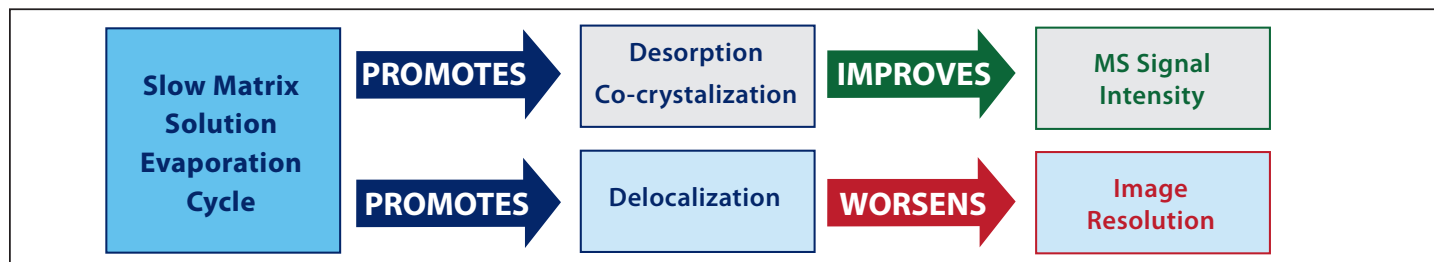
The proprietary spray technology of the HTX M5 Sprayer™ guarantees a very fine, uniform and consistent matrix coating crucial for high-resolution imaging and relative quantification of analytes.

The unique ability to control liquid and propulsion gas temperature creates a fine solution mist that can be deposited in a precise and adjustable pattern over all or part of any MALDI plate.

Spray characteristics (wet or dry) are easily adjustable via the intuitive operator interface. Users can create and save methods for reproducible operation.

Key Characteristics

- ◆ Proprietary technology providing very small matrix droplets (<5 microns)
- ◆ High flow rate and fast sample prep (2 to 18 minutes per slide)
- ◆ Highly consistent matrix deposition across entire sample area (+/- 3% by weight)
- ◆ Unique use of temperature and nitrogen flow to control evaporation rate and matrix crystal formation
- ◆ More than 30 validated protocols covering trypsin and most matrices (e.g.: SA, CHCA, DHB, DAN, 9-AA, DHA, CMBT, THAP)
- ◆ Validated protocols for Trypsin digestion of FFPE
- ◆ Continuous matrix coverage as needed for high-resolution imaging
- ◆ Rugged operation and easy clean-up



MATRIX DEPOSITION CHALLENGE

Addressing the Matrix Deposition Challenge

The main challenge when preparing samples for MALDI Mass Spectrometry Imaging is to balance the positive effects of the matrix solution penetrating the tissue and co-crystallizing with the analytes, and the negative effects of analytes delocalization.

The HTX M5 Sprayer™ builds on the proven capabilities of its predecessor, the HTX TM-Sprayer™, to offer even greater performance. The all-new M5 chassis, high velocity stage and heated sample holder drawer

contribute to a greater user experience and expanded process capabilities including:

- ◆ Faster and drier deposition capability
- ◆ On-tray trypsin digestion capability (with optional humidity chamber)
- ◆ On-tray sample re-hydration (with optional humidity chamber)

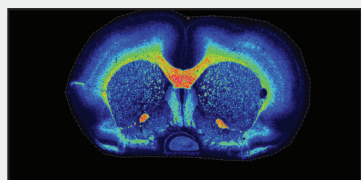
On the next page view a comparison of the two models.

SUCCESSFUL SPRAY DEPOSITION OF 2,4,6-TRIHIDROXYACETOPHENONE MALDI MATRIX

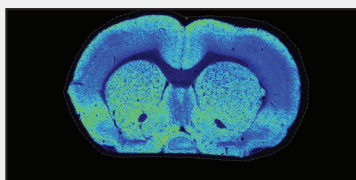
2,4, 6 Trihydroxyacetophenone or THAP offers great potential for MALDI analysis and Imaging of lipids, acidic glycans and glycol-peptides in negative ion mode, but its deposition by spray is challenging due to the tendency of THAP to form hydrates. When deposited by spray under normal conditions, the hydrated solution results in a liquid film layer that dries very slowly and forms a non-uniform coating. In the example below we devised a two steps process to deposit THAP successfully and create high resolution MS images.

Taking advantage of the high velocity of the XY stage of the HTX M5 Sprayer, the first four layers are deposited in super dry mode to create a seeding layer, then the next four layers, better optimized for extraction, are deposited on top resulting in a very good balance of analytes extraction, high signal intensity and limited spatial delocalization. This protocol did not require rehydration to produce a high intensity spectrum.

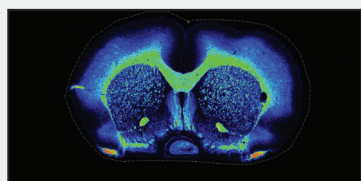
Data and images courtesy of Dr. Junhai Yang, Vanderbilt University.



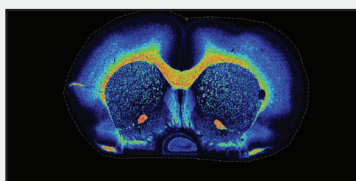
m/z 806.3 +/- 0.5Da



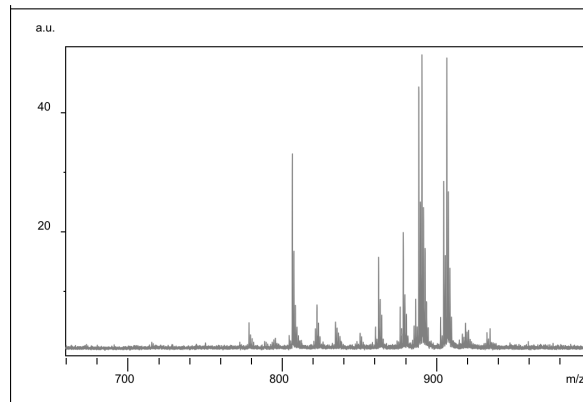
m/z 885.3 +/- 0.5Da



m/z 888.4 +/- 0.5Da



m/z 906 Da +/- 0.5 Da



PROTOCOL

First 4 passes:

Matrix: 2,4,6-THAP at 1mg/ml in 100% Acetone
Flow Rate: 0.050 ml/min
Velocity: 2,000 mm/min
Temperature: 30°C
Track Spacing: 1.5 mm

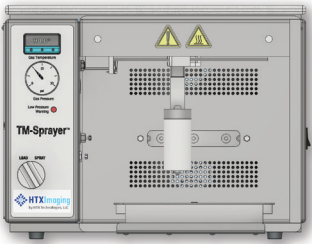
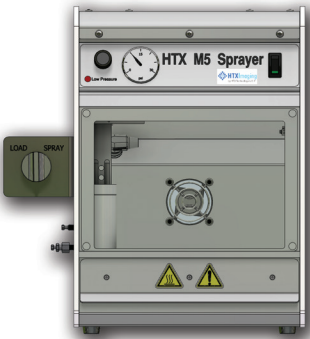
Next 4 passes:

Matrix: 2,4,6-THAP at 11.1mg/ml
in 66% Chloroform, 34% Methanol
Flow Rate: 0.100 ml/min
Velocity: 3,600mm/min
Temperature: 30°C
Track spacing: 1.5 mm

INSTRUMENTS

Sample Prep: Prototype HTX M5 Sprayer
Analysis: Bruker rapifleX TissueTyper
Acquisition raster: 15µm

HTX MALDI Sample Preparation Systems: M3 vs. M5

Product Name Front view	HTX TM-Sprayer	HTX M5 Sprayer
Chassis Instrument footprint Weight Available spray area Drawer loading Heated tray Waste/Drain	 M3 chassis 16.5"W x 13"H x 13"D 40 lbs. (18Kg) 180 x 140mm No No Cup	 M5 chassis 11.5"W x 16.3"H x 14"D 35 lbs. (16Kg) 180 x 140mm Yes Yes Trough gutter EZ in/out
XY stage Maximum stage velocity Nozzle height	1,350 mm/min Fixed (40mm)	5,600 mm/min Adjustable (25, 40, 52mm)
Heated Spray Nozzle	24V 40W RED Nozzle	Same
Electronics/Software Location COM channels	Left panel 2 COM Ports XY, Temperature	Top panel 4 COM Ports XY, Temp1, Temp2, Pump
Software version	TMSP CS 4.1	TMSP CS 5.1
Protocols	Methods.xml file offering 30+ validated protocols	Expanded but fully compatible with all previous methods.

HTX TECHNOLOGIES PRODUCTS ARE FOR RESEARCH USE ONLY. NOT FOR USE IN DIAGNOSTICS PROCEDURES.

HTX M5 Sprayer™ is available worldwide exclusively from HTX Technologies, LLC.
To request further information contact:

Alain Creissen

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acreissen@htximaging.com

HTX Technologies offers innovative sample preparation systems for advanced analytical platforms. Our integrated workflow solutions include user training, instruments, software, consumables and method development services.



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